MATHEMATICS DEPARTMENT
FAITH · RIGHTEOUSNESS · WISDOM

cognia SCHOOL OF DISTINCTION 2024

مدارس دار الفكر
DAR ALFIKR SCHOOLS

GRADE 10 · ALGEBRA II · TOPIC 2

Lesson 2-4

Overview of Complex Numbers

$$i^2 = -1, \quad z = a + bi$$

Journey of 6 stations, 60 minutes

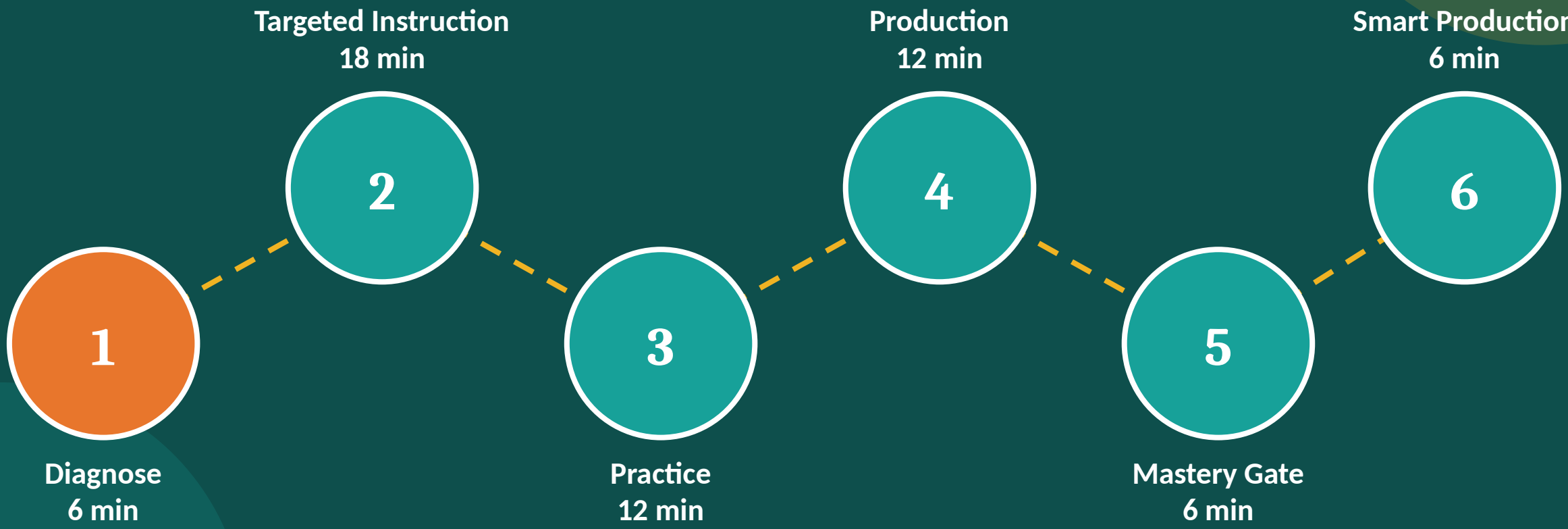
Quadratic Functions and Equations

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Today's journey

Six stations. Each one zooms in as we arrive.



What we are learning

OBJECTIVES

1. Demonstrate the role of the imaginary unit i in the system of complex numbers.
2. Evaluate expressions involving the imaginary unit i and complex numbers.
3. Use the properties of complex numbers to perform operations such as addition, subtraction, and multiplication.
4. Use complex conjugates to simplify quotients of complex numbers.
5. Solve real-world problems using complex conjugates.
6. Express the complex solutions of a quadratic equation in the form a + bi.

VOCABULARY

- Argument
- complex conjugate
- complex number
- complex plane
- imaginary axis
- imaginary part
- imaginary unit
- Modulus
- Norm
- purely imaginary number
- real axis
- real part

Standards

HSN.CN.A.1 Know there is a complex number i such that i2 = -1, and every complex number has the form a + bi with a and b real.

HSN.CN.A.2 Use the relation i2 = -1 and the commutative, associative, and distributive properties to add, subtract, and multiply complex numbers.

HSN.CN.A.3 Find the conjugate of a complex number; use conjugates to find moduli and quotients of complex numbers.

HSN.CN.C.8 Extend polynomial identities to the complex numbers.

HSA.REI.B.4.B Solve quadratic equations by inspection (e.g., for x2 = 49), taking square roots, completing the square, the quadratic formula and factoring, as appropriate to the initial form of the equation. Recognize when the quadratic formula gives complex solutions and write them as a ± bi for real numbers a and b.

STATION 1 · DIAGNOSE · 6 MIN

Start here: three quick checks

1

Simplify: $\sqrt{72}$

2

Expand: $(x+3)(x-2)$

3

Discriminant of x^2+2x+5

Done when

All three answers are written without a calculator, and you can say what a negative discriminant tells you about the real solutions.

STATION 2 · TARGETED INSTRUCTION · 18 MIN

The imaginary unit i

$$i = \sqrt{-1}, \qquad i^2 = -1 \qquad i^1 = i,\ \ i^2 = -1,\ \ i^3 = -i,\ \ i^4 = 1$$

EXAMPLE A

Simplify: $\sqrt{-49}$

$$= \sqrt{49} \cdot \sqrt{-1}$$

$$= 7i$$

EXAMPLE B

Evaluate: i^{27}

$$27 = 4 \cdot 6 + 3$$

$$i^{27} = \left(i^4\right)^6 \cdot i^3$$

$$= 1 \cdot (-i) = -i$$

EXAMPLE C

Simplify: $\sqrt{-4} \cdot \sqrt{-9}$

$$= 2i \cdot 3i$$

$$= 6i^2$$

$$= -6$$

STATION 2 · TARGETED INSTRUCTION · 18 MIN

Add, subtract and multiply

$$(a+bi)\pm(c+di)=(a\pm c)+(b\pm d)i$$

Multiply: distribute, then $i^2=-1$

EXAMPLE A

$$(3+2i)+(5-6i)$$
$$=(3+5)+(2-6)i$$
$$=8-4i$$

EXAMPLE B

$$(4-i)-(2-3i)$$
$$=4-i-2+3i$$
$$=2+2i$$

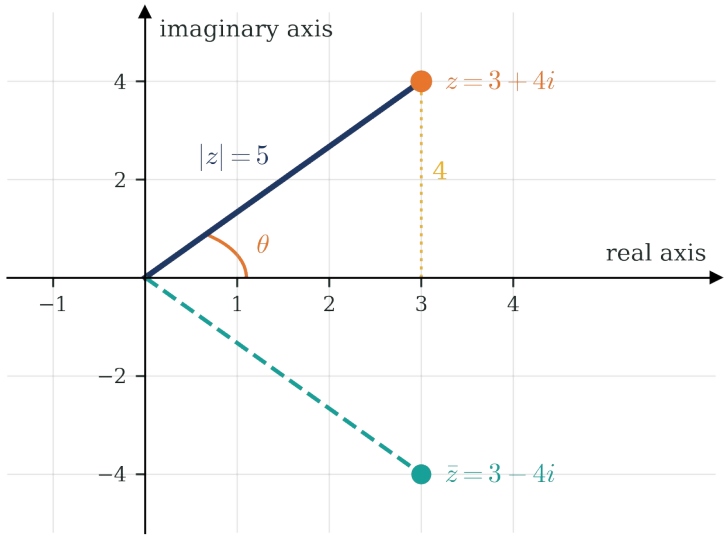
EXAMPLE C

$$(2+3i)(4-i)$$
$$=8-2i+12i-3i^2$$
$$=8+10i-3(-1)$$
$$=11+10i$$

STATION 2 · TARGETED INSTRUCTION · 18 MIN

Conjugates, modulus and the complex plane

$$\overline{a+bi} = a-bi \qquad (a+bi)(a-bi) = a^2+b^2 \qquad |a+bi| = \sqrt{a^2+b^2}$$



EXAMPLE A

$z = 3+4i$: find $z\bar{z}$ and $|z|$

$$z\bar{z} = (3+4i)(3-4i) = 9-16i^2$$

$$= 9+16 = 25$$

$$|z| = \sqrt{25} = 5$$

EXAMPLE B

Factor: x^2+9

$$x^2+9 = x^2-(3i)^2$$

$$= (x+3i)(x-3i)$$

STATION 2 · TARGETED INSTRUCTION · 18 MIN

Dividing and complex solutions

$$\frac{a+bi}{c+di} = \frac{(a+bi)(c-di)}{c^2+d^2} \qquad x = \frac{-b \pm \sqrt{b^2-4ac}}{2a}, \quad b^2-4ac < 0$$

EXAMPLE A

Simplify: $\dfrac{5}{2-i}$

$$= \frac{5(2+i)}{(2-i)(2+i)}$$

$$= \frac{5(2+i)}{5} = 2+i$$

EXAMPLE B

Simplify: $\dfrac{3+i}{1+i}$

$$= \frac{(3+i)(1-i)}{(1+i)(1-i)}$$

$$= \frac{3-3i+i-i^2}{2} = \frac{4-2i}{2}$$

$$= 2-i$$

EXAMPLE C

Solve: $x^2-4x+13=0$

$$b^2-4ac = 16-52 = -36$$

$$x = \frac{4 \pm \sqrt{-36}}{2} = \frac{4 \pm 6i}{2}$$

$$x = 2 \pm 3i$$

STATION 3 · PRACTICE · 12 MIN

Give it a go: Practice, Apply, Investigate

PRACTICE

Secure the method

Simplify: $\sqrt{-75}$

Multiply: $(3-2i)(1+5i)$

Done when: both answers are written in the form a + bi with no i squared left and every step shown.

APPLY

Use it in context

Write $\dfrac{2+3i}{1-2i}$ as $a+bi$

Solve: $x^2+6x+10=0$

Done when: the quotient is checked by multiplying back and both solutions are written as a + bi and checked by substitution.

INVESTIGATE

Find out why

Why is $(a+bi)(a-bi)$ always real?

Is $\sqrt{-4}\cdot\sqrt{-9}=\sqrt{36}$? Explain.

Done when: your argument uses i squared equals -1 in general terms, not just one numerical example.

STATION 4 · PRODUCTION · 12 MIN

Production: the NEOM desalination pump circuit

THE MODEL

At a NEOM coastal desalination plant (a Vision 2030 project on the Red Sea), engineers check the alternating current (AC) feeding a seawater pump motor. In AC circuits, resistance and reactance combine into a complex impedance Z measured in ohms, and the current is the voltage divided by the impedance. Two parts of the circuit are connected in series, so their impedances add. All numbers are model values chosen for this lesson, not real plant data.

$$Z_1 = 2 + 3i, \quad Z_2 = 1 + i \quad (\Omega)$$

$$V = 120 \text{ V}, \quad I = \frac{V}{Z_1 + Z_2}$$

1. Add the two impedances to find the total impedance Z in the form a + bi.
2. Use the complex conjugate of Z to write the current I in the form a + bi.
3. Find the size (modulus) of the current and decide whether a 30 A circuit breaker is enough.

DONE WHEN

- Z is found by adding real parts and imaginary parts separately.
- The current is written as a + bi after multiplying the top and the bottom by the conjugate.
- The modulus is found with the square root of a squared plus b squared and is labelled in amperes.
- The breaker decision is written in a full sentence and states that the numbers are model values.

Hint for solving: the size of a quotient is the size of the top divided by the size of the bottom, so you can check your modulus as 120 divided by the modulus of Z.

STATION 5 · MASTERY GATE · 6 MIN

Mastery Gate: three questions to pass

1

Evaluate: i^{50}

2

Multiply: $(2+i)(3-2i)$

3

Simplify: $\dfrac{4}{1+i}$

Pass mark

3 of 3 correct unlocks Station 6. Fewer than that: return to the slide for the question you missed, redo it, then re-take the gate.

STATION 6 · SMART PRODUCTION · 6 MIN

Smart Production: make one, then check it

CREATE

1. Write one quadratic equation with integer coefficients that has no real solutions.
2. Solve it on the back and write both solutions in the form a + bi.
3. Swap with a partner, solve theirs and check one solution by substitution.

Done when: your solution is complete and correct, the two solutions are conjugates, and your partner agrees with every step.

TIME TO CHECK

Exit ticket: write your answer on a sticky note and place it on the door.

Solve: $x^2 + 4x + 13 = 0$

Write the solutions as $a + bi$

Next lesson: 2-5 Completing the Square.

Exam connections

SAT

If $\dfrac{7+4i}{2-i} = a+bi,\ a,b$ real,

what is b? $\left(i=\sqrt{-1}\right)$

STEPS

1 $= \dfrac{(7+4i)(2+i)}{(2-i)(2+i)}$

2 $= \dfrac{14+15i+4i^2}{5} = \dfrac{10+15i}{5}$

3 $= 2+3i, \quad b=3$

GAT

Which quantity is greater?

A: i^{2026} B: i^{2028}

STEPS

1 $2026 = 4\cdot 506+2 \Rightarrow \mathrm{A} = i^2 = -1$

2 $2028 = 4\cdot 507 \Rightarrow \mathrm{B} = 1$

3 B is greater

SAAT

The solutions of $x^2-2x+5=0$ are

STEPS

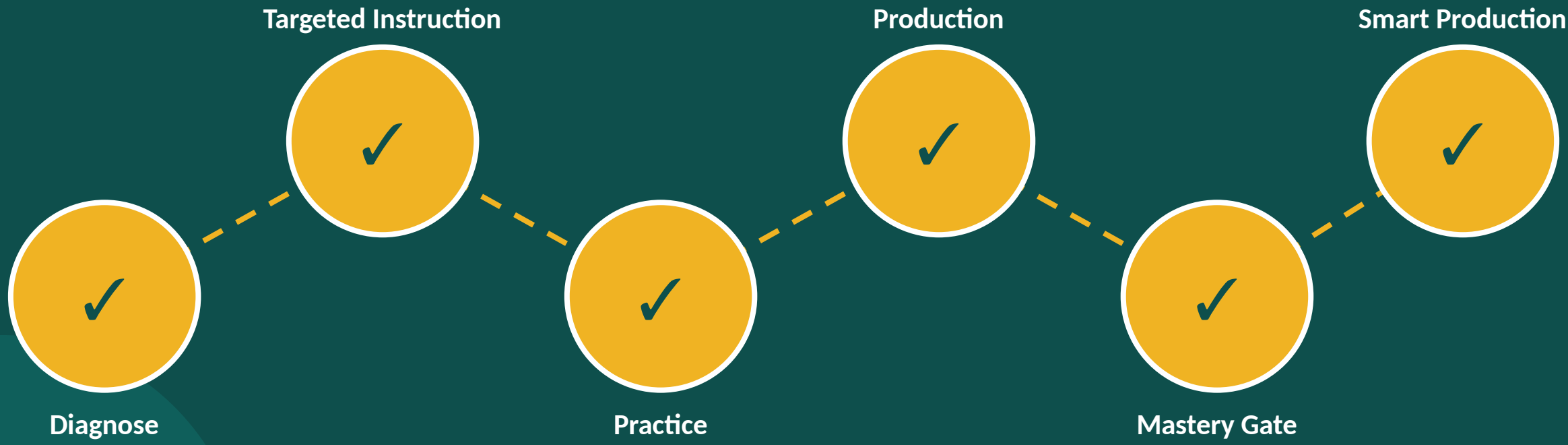
1 $b^2-4ac = 4-20 = -16$

2 $x = \dfrac{2\pm 4i}{2}$

3 $x = 1\pm 2i$

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Journey complete



i squared is -1: every complex number is a + bi, we add and multiply like binomials, divide with the conjugate, and every quadratic now has solutions.